

NANDHANA K S

Cybersecurity Researcher | Security Lab Developer | Learning Platform Builder

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PROFESSIONAL SUMMARY

Cybersecurity researcher and security lab developer with experience in web security, API security, SOC operations, threat intelligence, and security automation. Conducts security research and builds interactive learning platforms at Offenso Hackers Academy. Developed 50+ security labs, LMS platforms, and CTF environments used by 50+ students. Combines practical security knowledge with platform development (Python, Flask, React) and explores AI-augmented education, cloud security, and offensive security.

CORE COMPETENCIES

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|--------------------------|-----------------------|-----------------------|
| • Cybersecurity Research | • Web Security | • API Security |
| • SOC Operations | • Threat Intelligence | • Security Monitoring |
| • Cloud Security | • Digital Forensics | • Security Automation |
| • Incident Response | • Lab Development | • Linux |
| • Technical Training | • Platform Building | • Security Awareness |

PROFESSIONAL EXPERIENCE

Cybersecurity Researcher & Trainer Sep 2025 – Present

Offenso Hackers Academy, Calicut, Kerala

- Conduct cybersecurity research across web, API, cloud, and automation domains. Developed 50+ hands-on labs, simulations, and CTF challenges for skill development.
- Developed interactive learning platforms, cybersecurity simulations, and practical training environments to make cybersecurity education more visual, engaging, and concept-driven.
- Built multiple cybersecurity learning systems and hands-on labs that simplify complex security concepts through guided exploration, interactive demonstrations, and real-world scenarios.
- Built MODUS, a full-stack cybersecurity LMS with CTF, Linux simulator, gamification, RBAC, and analytics. Built WorkLedger for productivity tracking across 8+ training batches.
- Delivered training on offensive security, SOC operations, SIEM, threat intelligence, and VAPT to 50+ students. Mentor students through real-world attack/defense scenarios and vulnerability assessment.
- Designed and delivered cybersecurity awareness initiatives, workshops, and HackQuest summer bootcamp programs for under-18 students, introducing cybersecurity through practical demonstrations and interactive activities.
- Conducted cybersecurity awareness sessions focused on online safety, digital privacy, cyber threats, ethical hacking concepts, and responsible technology usage.

Cybersecurity Intern Jan 2025 | 15-Day Internship

Kerala Police Academy

- Conducted malware analysis using sandbox environments to study malicious behavior and identify threat indicators.
- Performed basic static and dynamic malware analysis techniques and documented findings.
- Developed a mini-project titled "Malware Analysis Using Sandbox Technology."

FEATURED PROJECTS

MODUS — Cybersecurity LMS React | Flask | SQLite | JWT

- Serves 128+ students across 6 cohorts with 24 CTF challenges, browser-based Linux simulator, gamification engine (XP/leaderboards), and analytics. Achieved 87% lab completion rate.
- Features RBAC (Admin/Instructor/Student), examination engine, notification system, and structured learning modules.

WorkLedger — Productivity Platform Flask | SQLite | Analytics

- Tracks 3,000+ tasks and 500+ hours across 8+ batches with productivity analytics, automated reporting, and batch monitoring.

OWASP Top 10 Interactive Lab Web Security | Interactive Learning

- Covers SQL Injection, XSS, and Broken Authentication with visual explanations and guided exploitation in a controlled environment.

Threat Intelligence Aggregator Python | VirusTotal | AbuseIPDB | Shodan

- Automates IOC enrichment from multiple threat intelligence sources for security operations and threat analysis workflows.

RESEARCH INTERESTS

- Web Security
- API Security
- Cloud Security
- Security Automation
- Threat Intelligence
- AI in Cybersecurity
- Security Education Technologies

TECHNICAL SKILLS

Security Tools: Burp Suite, Nmap, Wireshark, Metasploit, Nessus, Nikto, SQLmap, Gobuster, Hydra, John the Ripper, Kali Linux

SOC & Monitoring: Wazuh SIEM, Splunk SIEM, Threat Intelligence, Incident Response,

Programming: Python, Flask, FastAPI, Streamlit, HTML, CSS, JavaScript, React, Node.js, SQLite, Git, Docker

Platforms & OS: Kali Linux, Ubuntu, Windows, Linux Administration

Cloud & Identity: Microsoft Entra ID, Defender, Conditional Access, IAM, SC-900, Azure Fundamentals

CERTIFICATIONS

Professional Certifications

- Microsoft SC-900 — Security, Compliance & Identity Fundamentals
- Palo Alto Networks — Security Operations Fundamentals
- Cisco — Introduction to Cybersecurity
- EC-Council — Ethical Hacking Essentials (EHE)
- EC-Council — Network Defense Essentials (NDE)
- EC-Council — Digital Forensics Essentials (DFE)

Professional Learning Programs

- NPTEL — Systems and Usable Security
- NPTEL — Privacy and Security in Online Social Media
- IIT Kanpur — Cyber Security Premises Hacking Program
- Infosys Springboard — Incident Response & Forensics Program

Industry Simulations

- Deloitte — Cybersecurity Job Simulation
- Mastercard — Cybersecurity Job Simulation
- Tata — Cybersecurity Analyst Job Simulation

EDUCATION

Master of Computer Applications (MCA) Pursuing
Indira Gandhi National Open University (IGNOU)

B.Sc. Digital & Cyber Forensic Science 2022 – 2025
Bharathiar University, Tamil Nadu | CGPA: 8.3 / 10

PORTFOLIO & LINKS

Portfolio: <https://nandhanaks2004.github.io/portfolio/>

GitHub: <https://github.com/nandhanaks2004>

LinkedIn: <https://www.linkedin.com/in/nandhana-k-s-49b79827b/>